11 Building an AI RAG agent: Project walk-through

This chapter covers

- Building RAG apps with Next.js, Clerk, and LangChain.js
- Securing documents and isolating user data
- Using Upstash Redis and Vector for search and state
- Designing chat interfaces with text-to-speech
- Optimizing databases and prompt engineering
- Solving state sync and document-parsing issues

This project is a more complex full-stack generative AI web application: a retrieval-augmented generation (RAG) app designed to manage multiple knowledge bases. The app uses the Vercel AI SDK and LangChain.js libraries to enable users to upload documents, organize them into knowledge bases, and interact with these knowledge bases via conversational AI. The application also integrates modern authentication, persistence, and vector search technologies to ensure a more robust experience.